WAI YAN HPONE LAT (TEDDY)

APPLIED AI & ANALYTICS STUDENT • FULL-STACK & SOFTWARE ENGINEERING INTERN CANDIDATE

Nanyang Polytechnic
School of Applied Science & InfoTech

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🎯 **Full 44-Week Technology Internship (29 March 2027 – 28 January 2028):**
Year 2 Applied AI & Analytics student at Nanyang Polytechnic seeking a full-stack software engineering or applied data/AI internship. Proven track record architecting production-deployed React applications, Python and Node.js REST APIs, relational schemas with strict ACID integrity, and machine learning pipelines. Every project is verified with automated tests, cryptographic payment protection, and public cloud deployments.

CORE TECHNICAL CAPABILITIES

100+ Automated Tests Passing

💻 Frontend & Mobile Architecture

React 18 / 19 | **Vite** | **TypeScript** | **Tailwind CSS**
PWA & Service Workers | Web Speech API | Responsive UI/UX
State Management

⚙️ Backend, Serverless & Cloud APIs

Node.js | **Express.js** | **Python 3.12** | **FastAPI** | **Django REST**
Vercel Serverless | RESTful Architecture | OAuth 2.0 PKCE

🗄️ Relational Databases & Security

PostgreSQL (Neon) | **MySQL** | **Sequelize ORM** | SQLAlchemy | SQLite
Row-Level Locks | SSRF Hardening | CSRF & JWT Auth | RBAC

🤖 Applied AI, ML & Analytics

Google Gemini API | **scikit-learn** | **pandas / NumPy** | MLflow
LightGBM | KNIME Analytics | SAS Viya | Demand Forecasting

🚀 Payments, DevOps & Engineering Rigor

Stripe Checkout & Elements | **Signature-Verified Webhooks** | **Idempotent Event Ledgers** | Git & GitHub Actions | Vercel Cloud Hosting
Node Test Runner / Vitest / Pytest | Postman | Zero Regressions Protocol

VERIFIED ENGINEERING PROJECTS DIRECTORY

Live Deployments & Public Source

Project Name	Core Technologies	Key Architectural Innovation	Verified Status
GameVault	React 18, Express, MySQL, Stripe, Vercel	Server-owned pricing, idempotent webhooks, cart drawer	Live Storefront
Antigravity Mobile Coder	React PWA, Gemini API, Voice Speech, Node	Voice-driven PWA, self-healing test loops, multi-key pool	Live PWA App
SOLE DISTRICT	React 19, Django, PostgreSQL, Stripe	Row-level SKU stock locking, AI styling assistant, Neon DB	Live Storefront
SCENTHAUS Intelligence	FastAPI, scikit-learn, MLflow, Stripe	Content-based scent cosine similarity & demand forecasting	Verified System
MeterWise	TypeScript, FastAPI, LightGBM, Recharts	Singapore HDB estate energy analytics & anomaly detection	Verified System
Playlist Port	Node.js, Express, Spotify Web API, OAuth 2	OAuth 2.0 PKCE, server-held tokens, paginated track cloning	Verified System

💡 **Engineering Integrity:** All featured software is strictly verifiable via public repositories, comprehensive automated test suites, cryptographic boundary protection, and live URLs.

Deployed • 76 Automated Tests Passing

GameVault — Digital Game Subscriptions & Gift Cards Marketplace

Live Storefront | GitHub Repo

React 18 • Node.js • Express.js • MySQL • Sequelize ORM • Stripe Checkout • Vercel Serverless

System Overview: GameVault is a production-deployed digital gaming marketplace featuring 37 verified products across 15 game subscriptions (Xbox Game Pass, PlayStation Plus, Nintendo Switch Online, EA Play), 8 digital gift cards, and 14 digital games. Engineered with an interactive sliding cart drawer, real-time promo discount math (`GAMER10`), server-authoritative pricing, cryptographic Stripe webhook verification, and a staff fulfillment queue.

76/76 Automated Tests | **37** Products Live | **Zero** Client Price Tampering | **< 2.4s** Production Build

Core Architectural Innovations:

- **Server-Authoritative Price Validation:** Protects against browser client tampering. Cart items, quantities, and discounts are recomputed server-side from MySQL records before initiating Stripe Checkout sessions.
- **Interactive Sliding Cart Drawer:** Features quantity clamps (1..5), instant subtotal updates, dynamic promo code evaluation, and seamless outside-click dismissal.
- **Cryptographic Stripe Webhook Pipeline:** Signature-verified webhook processing with an idempotent ledger table preventing duplicate order fulfillments on Stripe event replays.
- **Staff 1-Hour Manual Fulfillment Queue:** Dedicated operations queue for email delivery of digital codes, tracking customer contacts, regional variants, and completion timestamps.
- **Edge Serverless Functions:** Deployed high-throughput Vercel edge endpoints (`/api/checkout/sessions`, `/api/promos/validate`, `/api/health`).

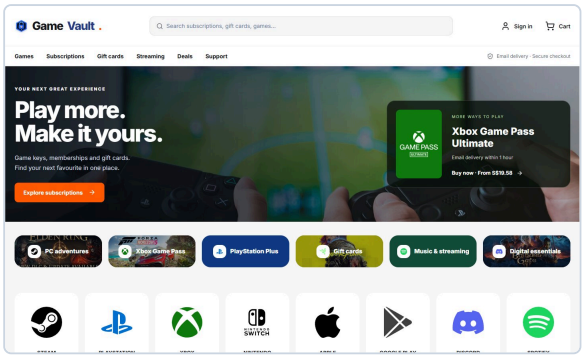


Figure 1.1: Live GameVault storefront on Vercel displaying verified subscription services, direct brand-tile routing, and curated deals.

Security & Concurrency Safeguards:

• Strict anti-tampering guards reject live checkout creation for unstocked or demo listings.
• 56 backend tests verify SQL transactions, token expiration, and CSRF defense headers.
• 20 frontend tests verify promo code calculations, cart bounds, and route deduplication.

End-to-End System Architecture & Transaction Lifecycle:

Client UI (React 18 + Cart Drawer) → Vercel Serverless (/api/checkout) → Stripe Hosted Checkout → Signed Webhook (Idempotent Ledger) → Staff Fulfillment Queue

Deployed • 22 Automated Tests Passing

Antigravity Mobile Coder — Autonomous Mobile Engineering PWA

🔗 Live PWA App | 📦 GitHub Repo

React • Progressive Web App (PWA) • Google Gemini Interactions API • Web Speech API • Vercel Edge

System Overview: Antigravity Mobile Coder is an installable Progressive Web App (PWA) that frees software engineers from desktop workstations. Developers command autonomous AI coding workflows from mobile phones using one-tap voice dictation, multi-key Google Gemini API pools, self-healing test execution loops, and SSRF-hardened cloud browser engines.

22/22 Automated Tests | **100%** Mobile PWA Score | **Multi-Account** Key Pool | **SSRF** Cloud Defense

Core Architectural Innovations:

- **Hands-Free Web Speech API Voice Dictation:** Captures developer intent naturally on mobile, appending dictated instructions without overwriting drafts or triggering keyboard auto-zoom.
- **Automated Multi-Key Rotation Pool:** Automatically detects HTTP 429 rate limits across pooled Google Gemini accounts, switching active keys instantly with live quota tracking.
- **Self-Healing Autonomous Test Loops:** Dispatches code generation into isolated sandboxes, executing `npm test` and `npm run build` repeatedly until exit code 0 is achieved.
- **Cloud Browser Engine & Session Vault:** Hardened serverless proxy with SSRF protection blocking loopback (`127.0.0.1`) and cloud metadata (`169.254.169.254`) IPs.
- **Laptop-Offline Cloud Dispatch:** Dispatches mobile-initiated issues directly to cloud runners even when developer workstations are powered off.

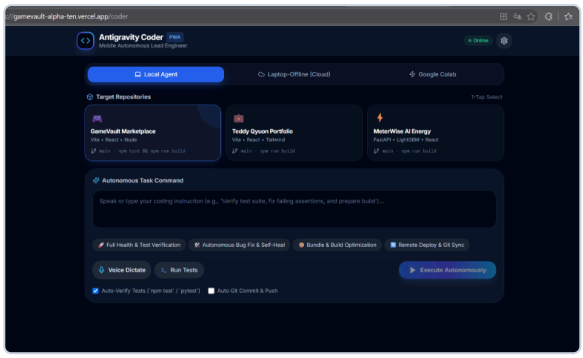


Figure 1.2: Antigravity Mobile Coder live PWA dashboard with voice dictation, repository presets, quota tracking, and live terminal output.

PWA & Engineering Quality Metrics:

• 22 unit & integration tests verifying PWA manifest shortcuts, offline caching, and SSRF.
• Service Worker precaches core application shells for immediate launch.
• Clean Vite production build in under 3 seconds; zero external tracking libraries.

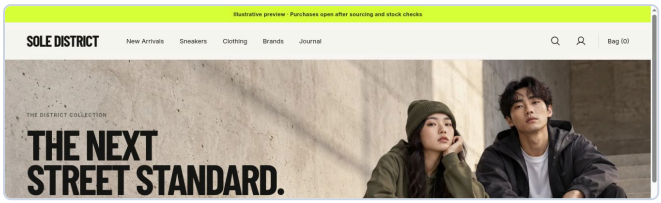
Autonomous Mobile Engineering Execution Pipeline:

Mobile Voice Prompt (Web Speech) → Gemini Interactions API Pool → Cloud Agent Sandbox Runner → Self-Healing (npm test / build) → Git Commit & Vercel Deploy

SOLE DISTRICT

Live

Streetwear Commerce & Django Workspace



An editorial streetwear commerce platform with a **Python/Django REST API** and dedicated **Neon PostgreSQL** database in Singapore. Built to solve size-level stock overselling.

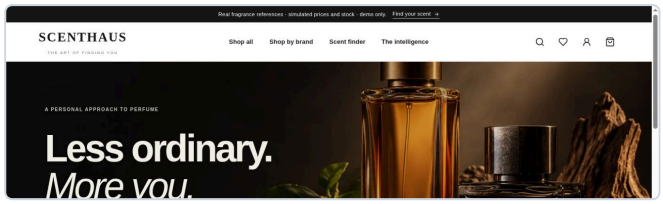
- **PostgreSQL Row Locks:** `select_for_update()` prevents concurrent overselling during checkout races.
- **Gemini 3.8 Flash Stylist:** AI assistant providing personalized outfit styling advice.
- **42/42 Backend CI Tests:** Verified on PostgreSQL 16 covering CSRF and role gates.

React 19 | Python | Django REST | PostgreSQL | Stripe

SCENTHAUS Intelligence

Verified

Fragrance Discovery & ML Recommendations



A luxury fragrance discovery platform with **35 perfume houses and 150 items**, combining a React frontend with a **FastAPI** ML recommendation pipeline.

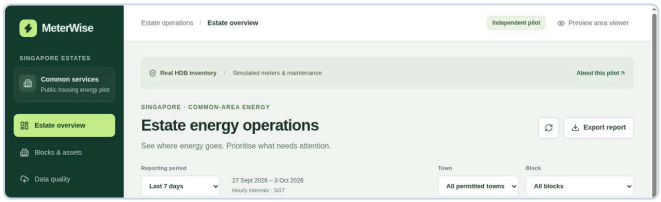
- **Scent Note Vectorization:** Cosine similarity recommendation engine over olfactive note profiles using `scikit-learn`.
- **Demand Forecasting ML:** Seasonal demand forecasting models tracked with `MLflow`.
- **FastAPI Microservice:** Asynchronous REST backend delivering sub-50ms responses.

React | FastAPI | scikit-learn | MLflow | PostgreSQL

MeterWise

Verified

Smart Estate Energy & Facilities Analytics



An IoT energy management dashboard built with **React/TypeScript** and **FastAPI**, utilizing public Singapore HDB estate data to monitor facility power usage.

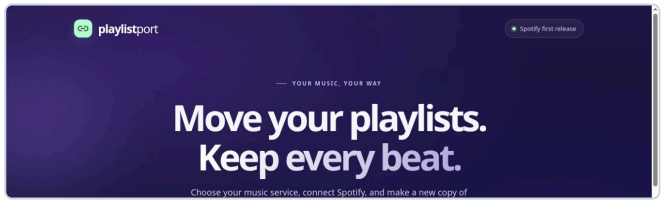
- **LightGBM Anomaly Detection:** ML models flag unexpected power surges and electrical faults.
- **Validated CSV Ingestion:** Automated schema validation sanitizing smart-meter batch uploads.
- **Interactive Timeseries Charts:** Built with Recharts with dynamic date-range downsampling.

TypeScript | React | FastAPI | LightGBM | SQL

Playlist Port

Verified

Spotify OAuth 2.0 Playlist Migration Utility



A full-stack utility enabling lossless playlist copying between Spotify accounts, solving API rate limits and token security issues through a secure server backend.

- **OAuth 2.0 PKCE Flow:** Server-held refresh tokens stored in HttpOnly cookies, shielding client.
- **Paginated Batch Processing:** Safely processes playlists with 1,000+ tracks via chunked recursive API pagination.
- **Deterministic Ordering:** Guarantees original curation sequence remains intact.

React | Node.js | Express.js | Spotify API | OAuth 2.0

NANYANG POLYTECHNIC COURSEWORK ENGINEERING

Grade 'A' Coursework Project Showcases

Annual Leave Management System — Enterprise Full-Stack System

Module Grade: A

Full Stack Application Development Module • Grade: A • Role: Full-Stack Contributor (Member 3)

A multi-role enterprise web application engineered with strict Role-Based Access Control (RBAC), hierarchical multi-tier approvals, and leave balance deductions.

- **Supervisor-to-Manager Approval Chain:** Implemented dynamic escalation routing applications based on organizational employee rank.
- **Atomic Quota Balance Deductions:** Guaranteed deductions execute strictly upon final authorization, preventing negative balances.
- **Acting Approver & Delegation Logic:** Architected temporary proxy appointment system with full audit trails for managers on leave.
- **Duplicate Submission Guards:** Built backend idempotency token guards preventing accidental double deductions caused by rapid client-side clicks.

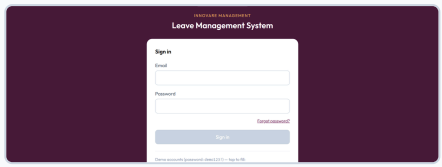


Figure 3.1: RBAC Sign-in & multi-role selection.

React | Express | Sequelize | MySQL

Gym Calories Predictive Machine Learning Analysis

Module Grade: A

Predictive Analytics & Forecasting Module • Grade: A • Tooling: KNIME Analytics & SAS Viya

End-to-end predictive modeling project comparing linear and ensemble machine learning algorithms to predict caloric expenditure during cardiovascular and resistance training workouts.

- **Data Cleansing:** Ingested 4,865 raw records; cleansed outliers and invalid heart-rate readings to produce 950 validated records.
- **Feature Engineering:** Constructed interaction variables between Workout Intensity, Mean Heart Rate, and Session Duration.
- **Algorithm Benchmarking:** Evaluated Linear Regression, Decision Trees, Random Forest, and Gradient Boosting on a standardized 70/30 train-test partition.
- **Model Selection:** Gradient Boosting delivered superior predictive accuracy (lowest RMSE and highest R^2), with Session Duration as the dominant driver.

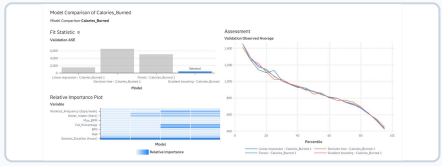


Figure 3.2: ML model comparison chart across splits.

KNIME | SAS Viya | Gradient Boosting

EDUCATION JOURNEY & ACADEMIC DISTINCTIONS

Verified Academic Record

Nanyang Polytechnic (Singapore) — Diploma in Applied AI & Analytics

2025 – Present (Expected May 2028)

Cumulative GPA: 3.18 / 4.00 • Year 2 Semester 1 GPA: 3.70 (Distinction Standing)

Key Coursework Highlights: Full Stack Application Development (Grade A), Predictive Analytics & Forecasting (Grade A), Data Wrangling (Grade A), Data Structures & Algorithms, Responsible AI for Sustainability, Database Administration, UX Design.

Myanmar Matriculation Exam

2018 / 2019

487 / 600 • 5 Distinctions

Distinctions in Mathematics, Chemistry, Physics, Biology, and Myanmar Language.

IBM SkillsBuild Credentials

Certified

Web Dev & UX Design Fundamentals

Standardized certifications in web development principles and user experience design.

WHY HIRE WAI YAN HPONE LAT (TEDDY)

High Execution Velocity & AI Literacy

Rapidly turns requirements into working software by leveraging modern AI agents, self-healing test loops, and automated build pipelines.

Security & Concurrency by Design

Enforces server-authoritative pricing, row-level locks, webhook idempotency, and SSRF defenses from day one.

100+ Automated Regression Tests

Commitment to reliability. Every flagship repository maintains comprehensive integration test suites guaranteeing zero regressions.

Full 44-Week Internship Availability

Committed for the complete 1-year cycle (29 March 2027 – 28 January 2028), ensuring sustained project ownership and engineering contribution.

Review Live Deployments & Verified Source Repositories:

Available for software engineering and full-stack technical interviews in Singapore.

Live Portfolio | GitHub Profile | Email Candidate